Abstract algebra – 2

Finite rings and division rings

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Theoretical minimum. This is the most frequent genre of the Romanian olympiad and a regular guest at the IMC: a finite ring is given together with one identity, and one has to recognise a field, a Boolean ring or a zero divisor.

Definitions. A *ring* is an abelian group $(A, +)$ with an associative multiplication distributing over addition; we always assume a unit $1 \neq 0$, but never commutativity. An element is a *unit* if it has a two-sided inverse, a *left zero divisor* if $ax = 0$ for some $x \neq 0$, *nilpotent* if $a^k = 0$, *idempotent* if $a^2 = a$. The *characteristic* is the least $n > 0$ with $n \cdot 1 = 0$. The *centre* is $\mathrm{Z}(A) = \{a : ax = xa \text{ for all } x\}$. A *division ring* is a ring in which every non-zero element is a unit; a commutative division ring is a *field*; a ring with $x^2 = x$ for all x is called *Boolean*.

Theorem 1 (finiteness turns injective into surjective). Let A be finite and $a \in A$. The maps $s_a(x) = ax$ and $d_a(x) = xa$ are additive maps of the finite set A into itself, so each of them is injective if and only if it is surjective. Hence every element of a finite ring is either a unit or a two-sided zero divisor, and *a finite ring without zero divisors is a division ring.* This one sentence solves about a third of all Romanian ring problems: whenever a hypothesis forbids a zero divisor, it produces an inverse for free.

Theorem 2 (Wedderburn). Every finite division ring is commutative, hence a field. Combined with Theorem 1: a finite ring without zero divisors is a field. The proof uses the class equation of the multiplicative group together with the inequality $|\Phi_n(q)| > q - 1$ for cyclotomic polynomials, and may be quoted.

Theorem 3 (characteristic). In a finite ring the characteristic n is finite and $n \cdot a = 0$ for every a. If A has no zero divisors the characteristic is a prime p; if $\mathrm{char}\, A = p$ then $(A, +)$ is a vector space over $\mathbb{F}_p$, so $|A| = p^k$. In a

commutative ring of characteristic p the map $x \mapsto x^p$ is a ring endomorphism (the *Frobenius*), because $\binom{p}{i} \equiv 0 \pmod{p}$ for $0 < i < p$.

Theorem 4 (identities forcing commutativity). A Boolean ring is commutative of characteristic 2: from $(x+x)^2 = x+x$ we get $2x = 0$, and from $(x+y)^2 = x+y$ we get $xy + yx = 0$, hence $xy = -yx = yx$. More generally (Jacobson) if $x^{n(x)} = x$ for every x with some exponent $n(x) > 1$, then the ring is commutative; olympiad problems use the special cases $x^2 = x$, $x^3 = x$ and $x^p = x$, which can be proved by hand.

Fact 1 (nilpotents and units). If a is nilpotent then $1-a$ is a unit, with inverse $1 + a + a^2 + \dots$ (a finite sum). In a commutative ring the nilpotent elements form an ideal; in a non-commutative ring this fails. For any a, b the element $1 - ab$ is a unit if and only if $1 - ba$ is, and the inverse is given by the identity $(1-ba)^{-1} = 1 + b(1-ab)^{-1}a$, which one may guess from the formal series $1 + ba + baba + \dots$

Fact 2 (idempotents split). If $e^2 = e$ then $1 - e$ is also idempotent, $e(1-e) = 0$, and in the commutative case $A \cong eA \times (1-e)A$ as rings. Hence a finite commutative ring with no non-trivial idempotents is local, and a Boolean ring with n elements has $n = 2^k$.

Fact 3 (finite fields). For every prime power q and every $n \geqslant 1$ there is an irreducible polynomial f of degree n over $\mathbb{F}_q$: the number of monic irreducible polynomials of degree n equals $\frac{1}{n}\sum_{d \mid n} \mu(d) q^{n/d} > 0$. Then $\mathbb{F}_q[x]/(f)$ is a field with q^n elements, and via the companion matrix of f it is realised inside $M_n(\mathbb{F}_q)$ as a set of q^n matrices in which every non-zero matrix is invertible. All fields with q^n elements are isomorphic.

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1. For a ring A and $a \in A$ define $s_a(x) = ax$ and $d_a(x) = xa$. (a) Prove that if A is finite then s_a is injective if and only if d_a is injective. (b) Give an example of a ring containing an element b for which one of s_b, d_b is injective and the other is not. (Румыния, 2014)

2. Let A be a finite ring and $a, b \in A$ such that $(ab-1)b = 0$. Prove that $b(ab-1) = 0$. (Румыния, 2018)

3. Prove that a Boolean ring is commutative of characteristic 2, and that a finite set with $n \geqslant 2$ elements can be equipped with the structure of a Boolean ring if and only if $n = 2^k$. Show also that the set of non-negative integers carries a Boolean ring structure.

(Румыния, 1998)

4. Let A be a finite commutative ring of prime characteristic p. Prove that $|A| = p^k$, that $x \mapsto x^p$ is additive, and that A has at most $|A|/p$ nilpotent elements. (Румыния, 2026)

5. Let A be a finite ring. Show that there exist natural numbers $m > p \geqslant 1$ such that $a^m = a^p$ for all $a \in A$. (Румыния, 2001)